UNIT – II

Design of Tension and Compression Members

Syllabus: Behaviour and Design of Simple and Built-Up Members Subjected to Tension – Shear Lag Effect – Design of Lug Angles – Tension Splice – Behaviour of Short and Long Columns – Euler's Column Theory – Design of Simple and Built-Up Compression Members with Lacings and Battens – Design of Column Bases – Slab Base and Gusseted Base.

What's inside these notes

- Tension members: modes of failure, net area, block shear
- Shear lag, design of angles, lug angles, tension splice
- Column behaviour, Euler theory, effective length
- IS 800 column curves (Perry–Robertson), buckling classes
- Design of simple compression members – worked example
- Built-up columns: lacing & battens
- Column bases: slab base and gusseted base

Exam tip: Tension: write the THREE checks (yield, rupture, block shear) and take the least. Compression: find $\lambda \rightarrow f_{cd}$ from $\chi \rightarrow P_d = f_{cd} \cdot A_e$. Always check the weaker axis.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2 \times 10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Tension Members – Behaviour

Members carrying **axial tensile force** – tie rods, hangers, bottom chords of trusses, bracing, cables.

Types: rods/bars (round, flat), single angle, double angle (back-to-back), channels, I-sections, built-up (lattice) members.

Behaviour

- Under tension the member **does not buckle** → full material strength can be used.
- Bolt holes reduce the area at the connection → **net section** fails first (stress concentration at hole).
- Ductile steel: gross section yields (large elongation) before rupture ; yielding at holes redistributes stress.
- Eccentric connection (angle bolted through one leg) → **shear lag** and bending.

Design strength of tension member (IS 800 Cl. 6) – least of

(1) Yielding of gross section: $T_{dg} = A_g f_y / \gamma_{m0}$

(2) Rupture of net section: $T_{dn} = 0.9 A_n f_u / \gamma_{m1}$

(3) Block shear: T_{db} (tearing of a block at the end connection)

$\gamma_{m0} = 1.10$, $\gamma_{m1} = 1.25$. Design: $T_u \le T_d = \min(T_{dg}, T_{dn}, T_{db})$.

Slenderness limits (IS 800 Table 3)

Member	Max KL/r
Tension, normally in tension only	400
Tension – may reverse (wind / seismic)	350
Tension – reversal due to other loads	180
Compression (DL + LL)	180
Compression (wind/EQ only)	250
Compression flange of beam	300

Slenderness limit prevents sag, vibration & damage in handling.

> Rods: use $A_n = 0.78 A_g$ for threaded part (net area at threads) – $T_d = 0.9 f_u A_n/\gamma_{m1} \le f_y A_g/\gamma_{m0}$.

2. Net Area, Staggered Holes, Block Shear

Net area of plate with holes

$$A_n = (b - n d_0) t$$

Staggered holes – take the critical zig-zag line:

$$A_n = [\,b - n\,d_0 + \Sigma\, p_s^2/(4\,g)\,]\,t$$

n = holes on the failure path, p_s = stagger pitch, g = gauge between rows. The path with the **least A_n** governs.

Angles – net area

- Connected leg: $A_{nc} = (a_1 - d_0 - t/2)\,t$ (a_1 = connected leg width)
- Outstanding leg: $A_{go} = (a_2 - t/2)\,t$
- Staggered: consider the unfolded angle (gauge $g = g_1 + g_2 - t$).

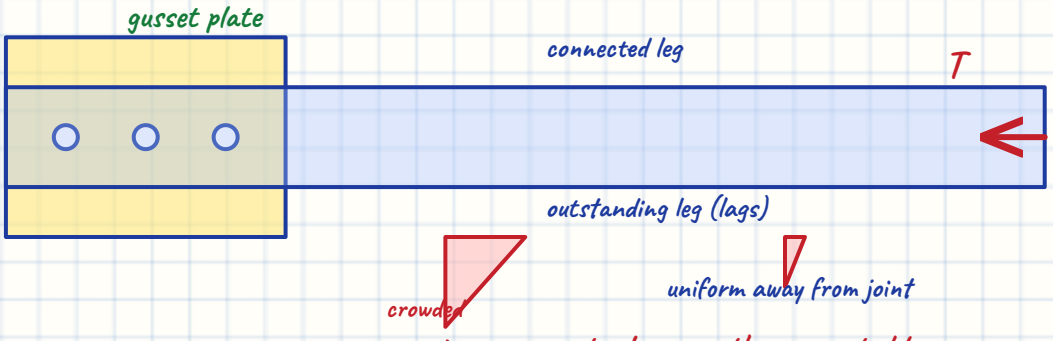
Block shear (IS 800 Cl. 6.4)

Failure by shear along bolt line(s) + tension across the end : a "block" tears out.

$$T_{db1} = A_{vg}\,f_y/(\sqrt{3}\,\gamma_{m0}) + 0.9\,A_{tn}\,f_u/\gamma_{m1}$$
$$T_{db2} = 0.9\,A_{vn}\,f_u/(\sqrt{3}\,\gamma_{m1}) + A_{tg}\,f_y/\gamma_{m0}$$
$$T_{db} = \text{smaller of the two}$$

A_{vg}, A_{vn} = gross, net area in **shear** along the bolt line ; A_{tg}, A_{tn} = gross, net area in **tension** (perpendicular to load) .

> Think: shear area **yields** while tension area **ruptures** (T_{db1}), or shear area ruptures while tension yields (T_{db2}). Use the smaller.



Shear lag in an angle connected through one leg

3. Shear Lag Effect

In angles / tees / channels connected by **only some elements** (e.g. one leg), stress is **not uniformly distributed** across the section near the joint – the connected part is overstressed, the outstanding leg "lags". Hence the **net section efficiency is reduced** and the rupture formula is modified.

Single angle with bolted/welded connection through one leg

$$T_{dn} = 0.9\,A_{nc}\,f_u/\gamma_{m1} + \beta\,A_{go}\,f_y/\gamma_{m0}$$
$$\beta = 1.4 - 0.076\,(w/t)(f_y/f_u)(b_s/L_c)\;;\; 0.7 \le \beta \le f_u\,\gamma_{m0}/(f_y\,\gamma_{m1})$$

- w = outstanding leg width ; b_s = shear lag width = $w + w_1 - t$ (bolted) ; w_1 = connected leg width
- L_c = length of end connection (outer bolt to outer bolt, or weld length)
- A_{nc} = net area of connected leg ; A_{go} = gross area of outstanding leg

For **welded**: $b_s = w$ (outstand) , L_c = weld length.

For **other sections** (channels, I): $T_{dn} = 0.9\, A_{nc}\, f_u/\gamma_{m1} + \beta\, A_{no}\, f_u/\gamma_{m1}$ … (same idea, see Cl. 6.3.3).

EXAMPLE 1 – STRENGTH OF ISA 90 × 90 × 8 IN TENSION

Angle (A_g = 1379 mm²) connected by **one leg** with **3 bolts M16 in a line, pitch 50 mm**, end distance 30 mm, edge distance 40 mm, d_0 = 18, Fe 410 (f_y = 250, f_u = 410).

1. Gross yield: $T_{dg} = 1379 \times 250 / 1.1 =$ **313.4 kN**

2. Net rupture (shear lag):

$A_{nc} = (90 - 18 - 4) \times 8 = 544$ mm² ; $A_{go} = (90 - 4) \times 8 = 688$ mm²

$w = 90$, $b_s = 90 + 90 - 8 = 172$, $L_c = 2 \times 50 = 100$ mm

$\beta = 1.4 - 0.076 \times (90/8) \times (250/410) \times (172/100) = 0.50 < 0.7 \rightarrow$ **$\beta = 0.7$**

$T_{dn} = 0.9\times544\times410/1.25 + 0.7\times688\times250/1.1 = 160.6 + 109.5 =$ **270.0 kN**

3. Block shear: $A_{vg} = 130\times8 = 1040$; $A_{vn} = (130 - 2.5\times18)\times8 = 680$; $A_{tg} = 40\times8 = 320$; $A_{tn} = (40 - 9)\times8 = 248$

$T_{db1} = 1040\times250/(\sqrt{3}\times1.1) + 0.9\times248\times410/1.25 = 136.5 + 73.2 = 209.7$ kN

$T_{db2} = 0.9\times680\times410/(\sqrt{3}\times1.25) + 320\times250/1.1 = 115.9 + 72.7 = 188.6$ kN

$T_d = \min(313.4, 270.0, 188.6) =$ **188.6 kN** (block shear governs)

4. Design of Tension Members (Procedure)

- 1. Factored tension T_u known. Assume $T_d \approx T_{dg} \rightarrow$ **A_g required = $1.1\, T_u / f_y$**; add 20-40 % for holes/shear lag (angles) → choose section from tables.
- 2. Check slenderness ≤ 350/400 using r_{min}.
- 3. Decide connection (bolts/welds); compute number & arrangement.
- 4. Check T_{dn} and T_{db} (with shear lag for angles); revise if $T_d < T_u$.
- 5. Check deflection/sag for long members (self weight) ; tack bolts for built-up.

Choosing sections

Section	Remark
Round rod	very light, bracings; threaded ends; no bending stiffness
Single angle	simple; eccentric → shear lag → reduce capacity
Double angles back-to-back	gusset between – concentric, used for truss chords/webs
Channels	good for larger loads, bolts through web
Built-up	laced / battened for long, stiff members

Built-up tension members

- Two or more components joined by **tack bolts / tack welds** at pitch ≤ **1000 mm** (tension), so the parts act together.
- Lacing / battens (like columns) – min. force 2.5 % ; slenderness of each part between tacks ≤ 50 (or 70 % of member).
- Tack bolts **do not** count for strength.

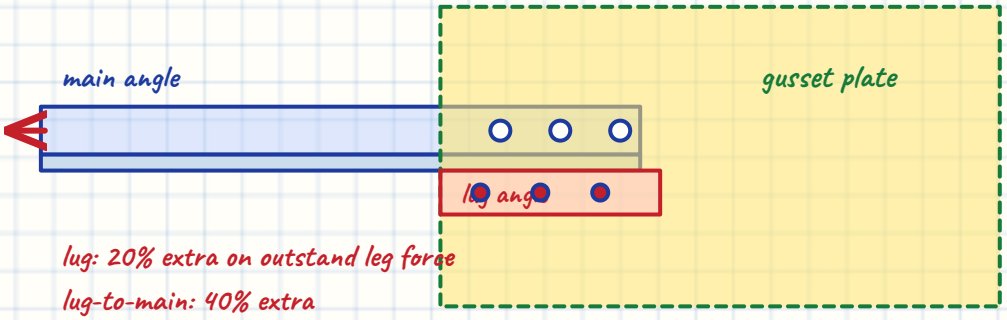
EXAMPLE 2 – REQUIRED AREA (QUICK)

T_u = 250 kN, bolted ISA pair, Fe 410.

$A_g(req) \approx 1.1 \times 250\times10^3/250 = 1100\ mm^2 \rightarrow \times1.3$ for holes/shear lag ≈ **1430 mm²**

Try **2 ISA 65x65x6** ($A = 2 \times 744 = 1488\ mm^2$) – then do three checks. Slenderness ($L = 2\ m$, $r_{min} \approx 12.6\ mm$) = 159 < 350 ✓.

5. Lug Angles



Lug angle – shortens the connection length

Problem: a long end connection in a channel/angle makes a big gusset. A **lug angle** (small angle) connects the **outstanding leg** to the gusset, so more of the section is connected → reduces length of connection & shear lag.

IS 800 requirements (Cl. 10.12)

- Lug angle + connection to gusset designed for **120 %** of the force in the outstanding leg of the main angle (20 % extra).
- Attachment of lug to main angle designed for **140 %** of that force (40 % extra).
- Effective length of the main member... lugs at least equal in thickness to the main angle; **at least 2 bolts/rivets** in the lug connection.
- Lug angles & main angle should be connected as near the end as possible.
- If lug angle is used the strength of the **member = full gross section** (no net area reduction beyond the hole line).

Lug angle design steps: Force in outstanding leg = $T \times (A_{outstand}/A_{total}) \rightarrow \times 1.2 \rightarrow$ bolts to gusset ; $\times 1.4 \rightarrow$ bolts/welds to main angle. Bolts shared by bending of the outstand.

6. Tension Splice

A splice joins two lengths of a tension member (rolled lengths limited to ~12 m).

- **Plates:** double cover plates (butt joint) – bolted or welded.
- **Angles:** cover angle (or plate) on both legs; **I-sections:** cover plates on flanges and web.

Design rule: the splice should carry the **full member strength** (or the design force) and be as strong as the member.

Steps (bolted plate splice)

- 1. Force to transmit $P_u = T_d$ of member (or actual).
- 2. Choose cover plate thickness so that $t_{cover} \geq t_{main}/2$ **each (double cover)** or total area ≥ main plate area.
- 3. Bolt value (double shear / bearing) → $n = P_u/V_d$ on each side of the joint.
- 4. Check cover plate for yield & rupture & block shear ; check length of lap.
- 5. Keep joint symmetric about the centroid → no eccentricity.

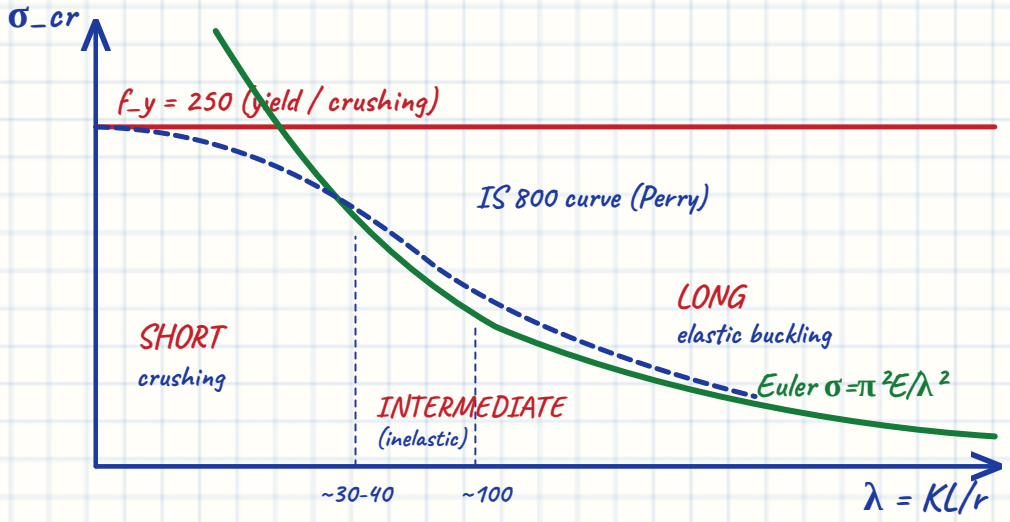
In flange splice of I-section: flange plates carry **M/d** (flange force) ; web plates carry **shear** (+ the web's share of moment).

7. Compression Members – Behaviour

Columns, struts, top chords of trusses, braces. Failure depends on **slenderness ratio** $\lambda = KL/r$ ($r = \sqrt{(I/A)}$).

- **Short column** (λ small, $\lesssim$ 30-40 for steel) – fails by **crushing/yielding** : $P = f_y A$.
- **Intermediate – inelastic buckling** (residual stresses, imperfection).
- **Long column – elastic buckling** at the Euler load (before yield).

Instability = loss of stiffness; deflection sidesway grows suddenly at P_{cr}.



Column strength vs slenderness

Factors that reduce the buckling strength

- Initial crookedness / eccentric load ; **residual stresses** (rolled & welded)
- End conditions (K) ; cross-section shape ; material non-linearity.

8. Euler's Column Theory

Assumptions: perfectly straight, centrally loaded, homogeneous, elastic (Hooke), small deflection, hinged ends, plane sections.

Derivation (hinged-hinged)

$EI\,y'' + P\,y = 0 \rightarrow y = A\sin(\pi x/L) \Rightarrow$

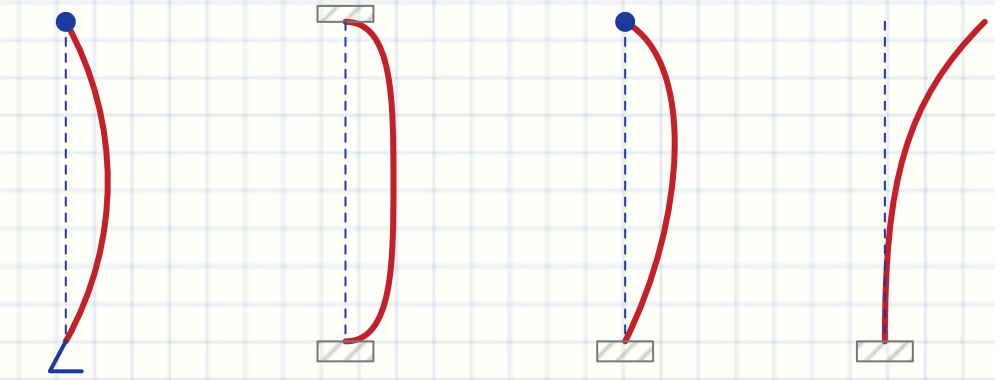
$$P_{cr} = \pi^2 E I / L^2 \text{ (buckles about weaker axis, } I_{min})$$

$$\sigma_{cr} = P_{cr}/A = \pi^2 E / \lambda^2,\ \lambda = L/r$$

With effective length $L_e = K L$

$$P_{cr} = \pi^2 E I / (K L)^2 = \pi^2 E I / L_e^2$$

Le = K · L (dotted = original, red = buckled shape)



Effective length for different end conditions

IS 800 Table 11 – effective length factors K

End conditions	K
Both ends held in position & fixed against rotation	**0.65**
Held in position at both ends, one end fixed, other hinged	**0.80**
Held in position, both ends hinged	**1.00**
One end fixed (held), other end fixed rotation but free to sway	1.20
One end fixed, other end hinged but free to sway	1.50
One end fixed, other free (cantilever)	**2.00**

Limitations of Euler's theory

- Valid only for **long columns** (λ > ≈ 100 for steel; σ_{cr} must be < proportional limit f_p).
- For $\lambda = \pi\sqrt{(E/f_y)} \approx$ **89** (f_y = 250), $\sigma_{cr} = f_y$.
- Ignores imperfections & residual stresses → real strength lower. Hence design codes use **Perry-Robertson curves** with imperfection factor α.
- Rankine formula (older): $1/P = 1/P_c + 1/P_E$.

9. Design Compressive Strength (IS 800 Cl. 7.1)

$$P_d = A_e \, f_{cd} \; ; \; f_{cd} = \chi \, f_y / \gamma_{m0} \le f_y / \gamma_{m0}$$

$$\chi = 1 / \left[\phi + \sqrt{(\phi^2 - \bar{\lambda}^2)} \right] \; ; \; \phi = 0.5 \left[1 + \alpha \, (\bar{\lambda} - 0.2) + \bar{\lambda}^2 \right]$$

$$\bar{\lambda} = \sqrt{(f_y / f_{cc})} = \sqrt{\left[f_y \, (KL/r)^2 / (\pi^2 E) \right]} \; ; \; f_{cc} = \pi^2 E / (KL/r)^2$$

α = imperfection factor ; A_e = effective area (full area for class 1-3).

Buckling class	a	b	c	d
α	0.21	0.34	0.49	0.76

Buckling class selection (Table 10, simplified)

Cross-section	Axis	Class
Rolled I ($h/b > 1.2$, $t_f \le 40$)	z-z / y-y	**a / b**
Rolled I ($h/b \le 1.2$, $t_f \le 100$)	z-z / y-y	b / c
Welded I ($t_f \le 40$)	z-z / y-y	b / c
Hot-rolled hollow (CHS, RHS)	any	**a**
Cold-formed hollow	any	b
Channels, angles, T, built-up from these	any	**c**

EXAMPLE 3 – ISMB 300 COLUMN, L = 4 M, HINGED BOTH ENDS

Properties: $A = 5626$ mm² ; $r_{yy} = 28.4$ mm (weak axis governs) ; $f_y = 250$; $E = 2 \times 10^5$.

$K = 1.0 \rightarrow \lambda = 4000/28.4 = \mathbf{140.8}$ (< 180 ✓)

$f_{cc} = \pi^2 E/\lambda^2 = 9.8696 \times 2 \times 10^5 \; / \; 140.8^2 = 99.6$ MPa

$\bar{\lambda} = \sqrt{(250/99.6)} = 1.585$

$h/b = 300/140 = 2.14 > 1.2$, $t_f = 13.1 < 40 \rightarrow$ y-y : **class b, $\alpha = 0.34$**

$\phi = 0.5[1 + 0.34(1.585 - 0.2) + 1.585^2] = 0.5[1 + 0.471 + 2.512] = 1.99$

$\chi = 1/[1.99 + \sqrt{(1.99^2 - 2.512)}] = 1/(1.99 + 1.206) = \mathbf{0.313}$

$f_{cd} = 0.313 \times 250 \; / \; 1.1 = \mathbf{71.1}$ MPa

$$P_d = 71.1 \times 5626 = 400 \text{ kN}$$

Design procedure of a simple column

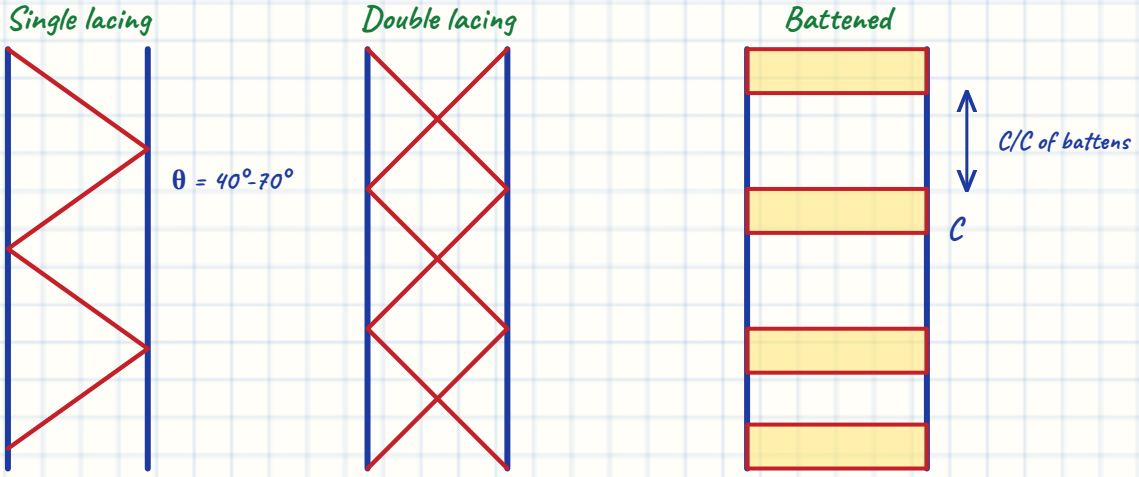
- 1. Assume $f_{cd} \approx 80$-120 MPa (for $\lambda \approx 60$-100) $\rightarrow A_{req} = P_u / f_{cd}$.
- 2. Select section (ISHB/ISMB, or built-up); get r_{min}.
- 3. Compute λ, buckling class, f_{cd} (Table 9 interpolation or formula), P_d.
- 4. If $P_d < P_u$ (or $\gg P_u$) revise. Check $\lambda \le 180$. Check local buckling (section class).
- 5. Design splice and base.

10. Built-up Compression Members

Used when rolled sections cannot give the required **area / radius of gyration** : two channels/ISMB back-to-back or toe-to-toe, box, 4 angles. Components are connected by **lacing** or **battens** (or continuous plates) so that they act as one unit.

Principle: keep $r_{zz} \approx r_{yy}$ (strong = weak axis) by spacing the components → "**effective radius**" larger.

- The **minimum distance** between channels is chosen so that $I_{yy} \geq I_{zz}$ (**equal strength about both axes**).
- Parallel axis theorem: $I_{yy} = 2\ [\ I_{own} + A\ (S/2)^2\ ]$.



Lacing (single / double) and battens

11. Lacing – Design Rules (IS 800 Cl. 7.6)

- Lacing = flats / angles / channels in zig-zag (**single**) or crossed (**double**) between the main components.
- **Angle of inclination θ** with the axis: **40° – 70°** (single), 40°–70° double.
- Slenderness of lacing bars **KL/r ≤ 145**. Effective length: L_e **= length between inner end bolts** (single, bolted) ; **0.7 × length** (welded, between welds) ; 0.7 L for double lacing too.
- Min. width of flat ≥ 3 × bolt dia ; **thickness t ≥ L_{lace}/40 (single) or /60 (double)** (L = distance between inner connections).
- **Transverse shear** V = 2.5 % of axial load in column (design lacing for V); shared equally by the lacing planes.
- Force in each lacing bar (single lacing, N planes): **F = V / (N sin θ)** ; for double lacing F = V/(2 N sin θ)
- Lacing is attached by bolts or welds; **tie plates** at ends of lacing.
- Main component slenderness between lacing connections $\lambda_1 \leq$ **50** and ≤ **0.7 × overall** λ of the member.
- **Effective slenderness ratio of laced column = 1.05 × λ_0** (λ_0 = max. actual slenderness).

Lacing: **battens at the ends** (tie plates) must be provided; lacing should not be mixed with battens in the same panel.

12. Battens – Design Rules (IS 800 Cl. 7.7)

- Battens = plates connecting main members rigidly, spaced along length – columns act like a **Vierendeel frame** (moments + shear in battens).
- Spacing C such that slenderness of **main component between battens ≤ 50** and ≤ **0.7 λ_0**.
- **Effective depth** of end batten ≥ distance between centroids of main components (a), intermediate batten ≥ 0.75 a ; **end batten ≥ 2 × width of one member**.
- Thickness t ≥ **1/50 × distance between inner edges of connections**.
- **Effective slenderness = 1.1 × λ_0** (more flexible than laced).
- At least **3 battens** per column; end ones as tie.

> Transverse shear $V = 2.5\ \%\ P_u$; longitudinal shear in batten $V_b = V\,C / (N\,S)$
> Moment in batten $M = V\,C / (2\,N)$

C = c/c spacing of battens along the length ; N = number of parallel planes of battens ; S = min. transverse distance between centroids of the connection groups.

Batten is checked for shear V_b, bending M, and its connection (bolts/welds) for shear + moment.

Procedure – design of laced/battened column

- 1. Select two channels/ISMBs for **area ≈ P_u/f_{cd}** (trial $f_{cd} \approx 100$). Check about the **minor axis** (y-y of each component) – this is a normal column check using **r_{min}**.
- 2. Choose spacing S so that **$I_{yy,total} \geq I_{zz,total}$** (usually clear spacing between toes ≥ 100 mm, ≥ 2 flange widths).
- 3. Compute effective slenderness (× 1.05 laced / × 1.1 battened) and **f_{cd}**; $P_d \geq P_u$.
- 4. Design lacing or battens ($V = 2.5\ \%\ P_u$) and connections.

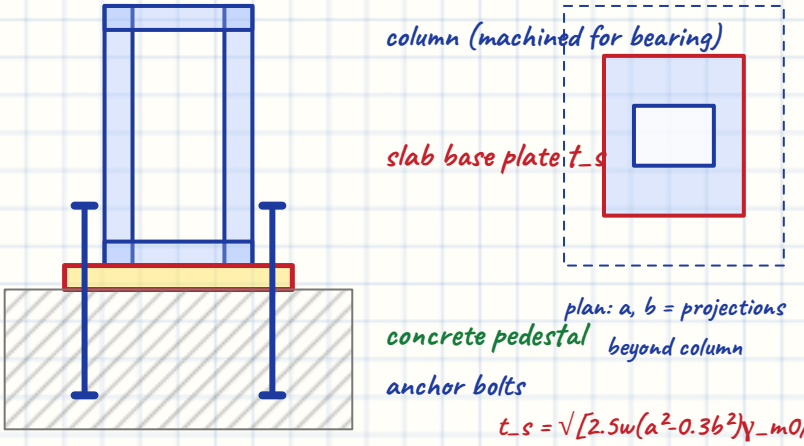
EXAMPLE 4 – LACING FORCE

Column carries P_u = 1200 kN with two lacing planes. V = 0.025 × 1200 = 30 kN → per plane = 15 kN.

Single lacing, θ = 45°: F = 15/sin 45° = **21.2 kN** in each lacing bar. Flat 50 × 6 : L_e (bolted) ≈ 250 mm ; r = 6/√12 = 1.73 mm → λ = 144 < 145 ✓ (OK, marginal) → use 50×8 for safety.

13. Column Bases – Slab Base

Column load P is spread over a large concrete area. Bearing strength of concrete (IS 456) = **0.45 f_{ck}** (confined: more).



Slab base – plate thickness from cantilever action

Design steps (IS 800 Cl. 7.4)

- 1. Required plate area $A_{req} = P_u / (0.45\, f_{ck})$. Choose B × L with projections beyond column ≥ 50–100 mm.
- 2. Bearing pressure $w = P_u/(B\,L) \le 0.45\, f_{ck}$.
- 3. Plate thickness (cantilever projection a (larger), b (smaller)):

$$t_s = \sqrt{[\,2.5\, w\, (a^2 - 0.3\, b^2)\, \gamma_{m0} / f_y\,]} \ge t_f \text{ of column flange}$$

- 4. If column end is machined for bearing, no weld needed to transmit load – **fillet weld** (tack) connection to the base plate (nominal) + **anchor bolts** (for erection, uplift, moment) ; 2–4 bolts of 20–24 mm.
- 5. Grout 25–50 mm below plate ; check shear at the base (friction + bolts).
- Plates up to **50 mm** are machined from a single slab; thicker (>50 mm) need machining of both faces.

> If bending of column exists, base has to resist **moment** → anchor bolts in tension, check bearing on one side.

EXAMPLE 5 – SLAB BASE FOR ISHB 300

P_u = 1000 kN, column ISHB 300 (h = 300, b_f = 250, t_f = 10.6), M25 concrete (f_{ck} = 25).

Permissible bearing = 0.45 × 25 = 11.25 N/mm² → A_{req} = 1000×10³/11.25 = 88 889 mm²

Provide **400 × 300** plate = 120 000 mm² → w = 1000×10³/120000 = **8.33 N/mm²** ✓

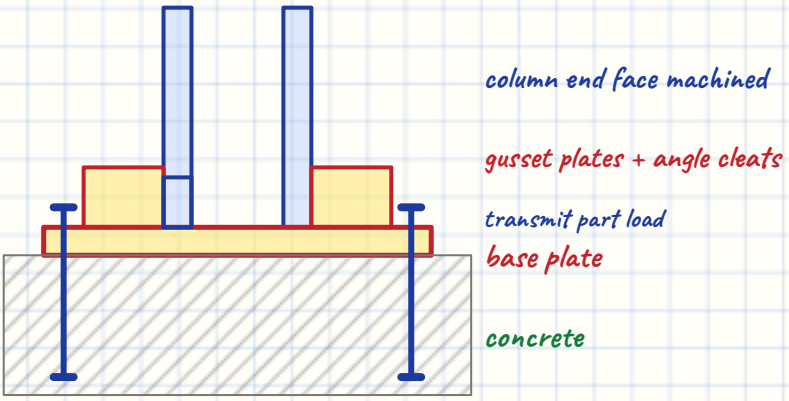
Projection: a = (400 − 300)/2 = 50 mm ; b = (300 − 250)/2 = 25 mm (choose a = larger, b = smaller)

$t_s = \sqrt{[\,2.5 \times 8.33 \times (50^2 - 0.3 \times 25^2) \times 1.1 / 250\,]} = \sqrt{(211.9)} =$ **14.6 mm** > t_f = 10.6 → provide **16 mm** plate.

Anchor bolts 4 No. M20 ; weld nominal 6 mm.

14. Gusseted Base

For **heavy loads** (large P or moments, where a plain slab would be too thick). Column end is **not** designed for bearing alone: gusset plates & cleats transfer the load.



Gusseted base

Components: column (flanges/web), **gusset plates** (on flanges), **angle cleats**, **base plate (thin, 20–30 mm)**, anchor bolts.

Design idea

- Load is shared: usually **assume 50–60 % through direct bearing** (machined column end on base plate) and rest via gussets and cleats.
- Gusset plate thickness ≥ web/flange of column, checked for bending as a cantilever carrying the portion of the load / pressure over its tributary width.
- Connection of column to gusset: bolts/welds for the load carried **by the gusset** (P × share).
- Base plate: designed as **cantilever** from face of gusset/angle, and as **continuous plate** between gussets, with $w = P/(B\ L)$:

$$M = w\,a^2/2 \text{ per unit width};\ t = \sqrt{(4\,M\gamma_{m0}/f_y)} \ldots (t^2 = 6M/(f_y/\gamma_{m0}) \text{ for rectangular section})$$

- Weld between gusset and base plate designed for the share of the load (+ moments).
- Anchor bolts for uplift (wind/moment).

Slab base vs Gusseted base

Feature	Slab base	Gusseted base
Load	light / medium	heavy
Plate thickness	thick (≥ 20 mm)	thin (gusset stiffened)
Fabrication	machined plate	welding & cleats
Cost	material high	fabrication high, lighter
Details	simple	complex

Other types : grillage footings for very heavy loads; moment-resisting bases with extended plates for portal frames.

> **Exam Q:** Always draw the base with column, base plate, grout, anchor bolts, concrete pedestal and label a, b, t, B, L.